

Test Plan — LearnWorlds Sandbox Course Purchase Flow

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1. Introduction

This document outlines the test strategy, scope, approach, and execution plan for validating the sandbox purchase flows on the LearnWorlds platform.

The objective is to ensure that users can successfully purchase courses through multiple checkout paths, including cart and direct purchase flows, with and without coupons, while maintaining system reliability, correct pricing, and a seamless user experience.

Important note: Since I am not 100% familiar with the business and functionality of learnworlds.com infrastructure, some assumptions will be taken in the context of this report that may not be 100% accurate for your platform.

Useful links:

- Cart & checkout flow can be enabled/disabled here:
https://pkalogerop.learnworlds.com/author/settings_ecommerce?tab=cart
- Sandbox payments are enabled and already pre-configured but can be customized here:
https://pkalogerop.learnworlds.com/author/payment_gateways

2. Objectives

- Validate successful course purchases via all supported flows
- Ensure accurate pricing calculations including coupons
- Verify checkout reliability in sandbox environment
- Confirm correct order creation and enrollment
- Detect regressions in purchase workflows
- Validate error handling and edge cases
- Ensure consistent user experience across flows

3. Scope

In Scope

- Purchase via cart flow
- Direct purchase (Buy Now / single course checkout)
- Coupon application scenarios
- Pricing calculations
- Enrollment after successful purchase
- Order confirmation
- Payment processing in sandbox
- Error handling
- UI/UX validation
- Basic security validations

Out of Scope

- Real payment gateway production testing
- Extensive automation testing, 2-3 cases will be selected only for automation
- Performance load testing
- Mobile testing
- Extensive security testing
- Third-party provider internal functionality
- Accounting and invoicing integrations

4. Test Strategy

Testing Levels

- System Testing
- Integration Testing
- End-to-End Testing
- Exploratory Testing
- Regression Testing

Testing Types

- Functional Testing
- Usability Testing
- Negative Testing
- Boundary Testing
- Compatibility Testing
- Security validation (basic)
- Data validation

Risk-Based Approach

Priority will focus on (from highest priority to lowest):

- Payment processing and Payment flow reliability
- Pricing accuracy and calculations
- Coupon logic correctness
- Enrollment correctness

Automation Strategy

Automation will target:

- Critical happy paths
- Regression purchase flows
- Coupon validation scenarios
- API validations for order creation

Tools for automation: Playwright with Typescript for UI tests / API with Typescript for backend tests

5. Test Coverage Matrix

Area	Coverage Level
Checkout flow	High
Pricing engine	High
Coupon logic	High
UI/UX	Medium
Notifications	Medium
Error handling	High
Security validation	Medium

High Coverage means: 75% and more covered tests cases

Medium coverage means: 50% to 75% covered test cases

Low coverage means: less than 50% covered test cases

6. Test Scenarios

Important note: each test scenario could be expanded more with additional Acceptance Criteria and details. Keeping it as simple as possible within the scope of this Test plan report.

6.1 Purchase Without Cart (Direct Checkout)

- Purchase single course successfully
 - Purchase with valid coupon
 - Purchase with expired coupon
 - Purchase with invalid coupon
 - Purchase without coupon
 - Payment failure handling
 - Retry payment
 - Cancel checkout
 - Session timeout during checkout
 - Verify enrollment after purchase
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6.2 Purchase With Cart Flow

- Add single course to cart and purchase
- Add multiple courses to cart and purchase
- Remove course from cart
- Update cart before checkout
- Apply coupon to cart
- Apply coupon to multiple courses
- Invalid coupon handling
- Cart persistence across sessions
- Checkout with empty cart

- Verify total price calculation
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6.3 Coupon Scenarios

- Percentage discount coupon
 - Fixed amount coupon
 - Coupon with usage limit
 - Coupon expiration validation
 - Coupon applied multiple times
 - Coupon stacking (if supported)
 - Coupon exceeding cart value
 - Coupon applied to specific course only
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6.4 Negative Scenarios

- Payment declined
 - Network interruption during checkout
 - Invalid payment details
 - Duplicate payment submission
 - Browser refresh during payment
 - Attempt to access checkout without login
 - Checkout with expired session
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6.5 Post-Purchase Validation

- Course enrollment confirmation
- Confirmation email triggered
- Order recorded in system

- Correct price recorded
- Coupon usage tracked
- Access to course content granted

7. Test Environment

- Sandbox payment gateway
- Production LearnWorlds environment (the optimal would be to have a staging or dev environment)
- Test user accounts
- Test courses
- Coupon test data
- Supported browsers. For the scope of this report only Chrome will be used. In the actual tests more browsers would have been used.

8. Test Data Strategy

Test data will include:

- Multiple pricing tiers
- Various coupon configurations
- Invalid coupon data
- Expired coupons
- Limited-use coupons
- Multiple user accounts

9. Entry Criteria

(note that since we are testing in production with predefined variables, the following will not be 100% accurate for the scope of this report)

- Sandbox environment available (testing in production for now)
- Test data prepared
- Test accounts created
- Feature deployed to test environment (testing in production for now)
- Requirements clarified. Acceptance criteria are present, accepted by all stake holders and well defined

10. Exit Criteria & Quality Gates

Exit Criteria:

- All critical scenarios executed
- No critical/high defects open
- Purchase success rate $\geq 99\%$
- Pricing calculations validated
- Test summary report completed

Quality Gates:

Release readiness requires:

- No critical defects
- $\geq 95\%$ pass rate
- Successful regression suite
- Stable checkout metrics
- Business approval

11. Risks & Mitigation

Important note: We could expand this analysis even further by applying ratios and percentage calculations, but we will go with the simple (likelihood * impact) formula here.

Risk	Likelihood	Impact	Score	Mitigation
Payment failures	Medium	High	High	Extensive negative testing
Pricing miscalculation	Medium	High	High	Pricing validation tests
Coupon logic bugs	High	Medium	High	Edge case testing
Environment instability	Medium	Medium	Medium	Retry strategy
Enrollment failures	Low	High	Medium	E2E validation

12. Observability & Monitoring Strategy

Post-release monitoring should track:

- Checkout failure rate
- Payment error rate
- Coupon error rate
- Enrollment failures
- Revenue anomalies
- Error logs

13. Defect Management

Defects will be tracked with severity levels:

- Critical — Payment failure / incorrect pricing
- High — Checkout blocked
- Medium — UI inconsistencies - Incorrect UI behavior
- Low — Cosmetic & Minor UX issues

Each defect will include steps to reproduce, logs, and environment details.

14. Deliverables

- Test plan
- Test scenarios
- Defect reports
- Test execution report
- Summary report
- Automation suite